

LECTURE 9.2

Turbulent Flow Prediction

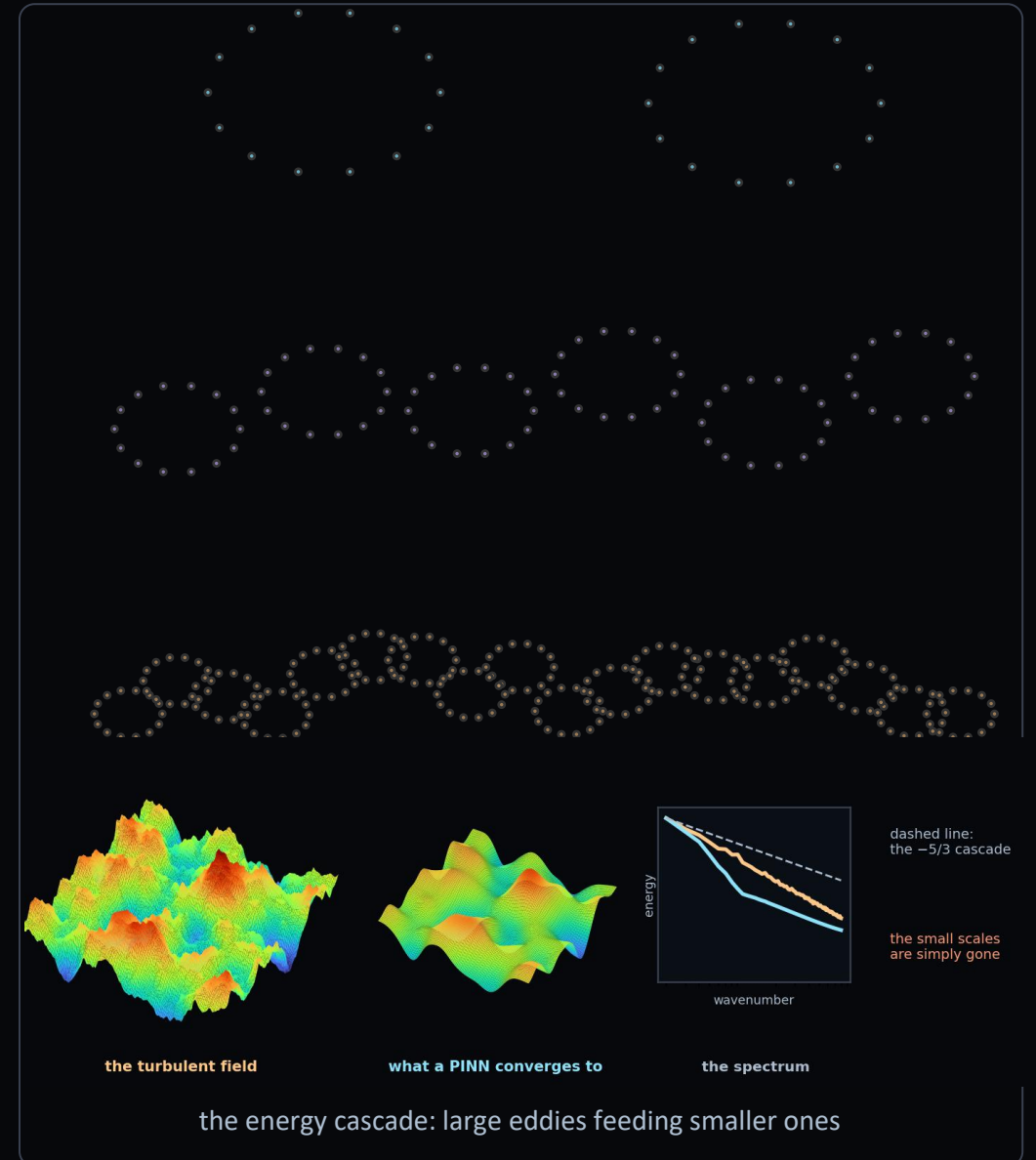
Turbulence is not solved, it is modelled. Reynolds averaging, the closure problem, and where a PINN genuinely contributes.

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Aalborg University · ret@et.aau.dk · Deep Learning for Engineering · 2026

Pope, Turbulent Flows (2000) · Raissi et al. (2019) · current PINN literature



What You Will Be Able To Do

THE GOALS, IN PLAIN WORDS

- **estimate** — the cost of resolving a turbulent flow, and say why it is prohibitive
- **derive** — where the Reynolds stress comes from, and why averaging leaves it behind
- **state** — the closure problem in one sentence
- **explain** — what an eddy viscosity assumes, and where that assumption fails
- **choose** — between prescribing a closure and learning one
- **argue** — why data assimilation, not forward solving, is the honest use case

WHAT TODAY ASSUMES

from L9.1	Navier-Stokes, and the Reynolds number
from L9.1	the stream function, and pressure
from L7.1	spectral bias

A LECTURE ABOUT LIMITS

Every other lecture asked how to solve an equation. This one asks what to do when you cannot, which is more useful than pretending otherwise.

AND ONE OPEN PROBLEM

Learning a closure from data is live research, not settled practice.

Why This Lecture Is Different

SAY THIS TO THE CLASS FIRST

A physics-informed neural network trained on the raw Navier–Stokes equations at high Reynolds number does not converge to the turbulent solution. It converges to something smooth — which is not a poor approximation of turbulence, it is not a solution at all.

- every previous lecture asked how to solve an equation
- **this one asks a different question** — what to do when you cannot
- turbulence is chaotic, multiscale, and three-dimensional even in a 2-D geometry
- **so engineering does not resolve it** — it averages the equations and models the rest

The L13 MiniProject returns to this material as an open research problem. Here we build the vocabulary for it.

What Makes Turbulence Hard

CHAOTIC

Sensitive to initial conditions. Two runs differing at the tenth decimal diverge into different instantaneous fields — though their statistics agree.

MULTISCALE

Energy enters at the largest scale and cascades down to the smallest, where viscosity dissipates it. The range of scales grows with Re .

THREE-DIMENSIONAL

Vortex stretching is the mechanism that sustains the cascade, and it does not exist in two dimensions. A 2-D 'turbulent' simulation is a different physics.

STATISTICALLY STEADY

The instantaneous field never repeats, but the mean, the variance and the spectra are reproducible. This is what makes engineering possible at all.

The last property is the one that saves us. Nobody needs the instantaneous field — an engineer needs the mean velocity, the pressure drop and the drag. Those are reproducible, and they are what we model.

The Arithmetic of Direct Simulation

SMALLEST TO LARGEST SCALE

$$\frac{\eta}{L} \sim Re^{-3/4}$$

- the smallest eddies shrink relative to the largest as Re rises
- so the point count grows as Re to the nine-fourths
- **at $Re = 10^4$** — water in a 5 cm pipe — that is far beyond any collocation set
- this is not a limitation of neural networks; it is why turbulence modelling exists
- **and a PINN has an extra handicap** — it is smooth by construction

POINTS NEEDED TO RESOLVE THEM ALL

$$N \sim Re^{9/4} \quad \text{per dimension in 3D: } Re^3$$

POINTS REQUIRED IN 3-D



$\log_{10}$ of the grid-point count, against Reynolds number

Split the Field, Average the Equations

REYNOLDS DECOMPOSITION: A MEAN PLUS A FLUCTUATION

$$u_i = \bar{u}_i + u'_i, \quad \overline{u'_i} = 0$$

AVERAGED CONTINUITY

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0$$

AVERAGED MOMENTUM — NOTE THE LAST TERM

$$\bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \nu \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} - \frac{\partial}{\partial x_j} \overline{u'_i u'_j}$$

- write every quantity as a mean plus a fluctuation, then average
- almost every term averages cleanly, because a fluctuation averages to zero
- the exception is the convection term, where two fluctuations multiply
- what survives is the Reynolds stress

- **note where it came from** — the same nonlinearity that made L9.1 hard

The Closure Problem

THE REYNOLDS STRESS TENSOR

$$\tau_{ij} = -\rho \overline{u'_i u'_j}$$

- the averaged equations contain more unknowns than equations
- deriving an equation for the Reynolds stress does not help
- it introduces triple correlations, and so on without end
- so the missing term must be modelled in terms of what you do have

THE CONSEQUENCE

unknowns > equations ⇒ closure required

COUNTING IN 3-D

mean velocities	3
mean pressure	1
Reynolds stresses	6
TOTAL UNKNOWNNS	10
equations available	4
SHORTFALL	6

no amount of algebra closes this gap

The Eddy-Viscosity Idea

BOUSSINESQ HYPOTHESIS — TURBULENCE ACTS LIKE EXTRA VISCOSITY

$$-\overline{u'_i u'_j} = \nu_t \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) - \frac{2}{3} k \delta_{ij}$$

EFFECTIVE VISCOSITY

$$\nu_{eff} = \nu + \nu_t$$

K-E MODEL

$$\nu_t = C_\mu \frac{k^2}{\varepsilon}, \quad C_\mu \approx 0.09$$

MIXING LENGTH

$$\nu_t = \ell_m^2 \left| \frac{\partial \bar{u}}{\partial y} \right|$$

- the oldest closure, and still the most used
- assume the Reynolds stress behaves like a viscous stress with a larger coefficient
- substitute back and RANS becomes Navier–Stokes with an effective viscosity
- so every tool from L9.1 applies unchanged
- **the catch** — the eddy viscosity is not a material property

The Wall Is Where It Gets Expensive

WALL COORDINATE

$$y^+ = \frac{u_\tau y}{\nu}, \quad u_\tau = \sqrt{\frac{\tau_w}{\rho}}$$

- next to a wall the mean velocity changes over a very short distance
- resolving it is where most of the cost sits
- **the saving grace is universality** — the log layer has the same shape for every wall flow
- classical codes exploit that with wall functions
- for a PINN it removes exactly the region needing the densest sampling

THE LOGARITHMIC LAYER

$$u^+ = \frac{1}{\kappa} \ln y^+ + B, \quad \kappa \approx 0.41, B \approx 5.0$$

THE UNIVERSAL PROFILE



viscous sublayer, then the log layer — the same for every wall

Solving RANS with a PINN

SUBSTITUTING THE CLOSURE

$$\nu_{eff} = \nu + \nu_t$$

- route one, pragmatic: prescribe the eddy viscosity from a standard model
- route two, interesting: represent it with a second network and learn it
- route two is where a PINN offers something a classical code cannot
- it is also open research rather than settled practice

OR LEARN THE CLOSURE ITSELF

$$\nu_t = \mathcal{N}_{\hat{\nu}}^t(\mathbf{x}) \geq 0$$

Prescribed ν_t

Standard model, PINN solves the mean flow.
Reliable. Exercise-ready.

Learned ν_t

Second network, constrained by residual and data.
Research territory. MiniProject.

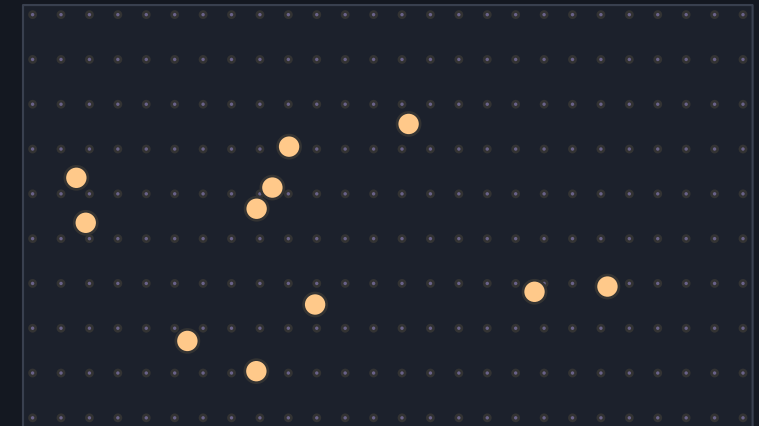
Data Assimilation Is the Honest Use Case

PHYSICS AND MEASUREMENTS IN ONE LOSS

$$\mathcal{L} = \mathcal{L}_{RANS} + w_d \mathcal{L}_{data}$$

- **turbulent flows are measured all the time** — PIV, hot wire, pressure taps
- a PINN fills the gaps under the constraint of the averaged equations
- so it is not competing with a CFD code at solving
- it is doing something the code cannot do at all
- **and it degrades gracefully** — data-driven where data is dense, physics-driven where it is not

SPARSE DATA, FULL FIELD



amber: sensors · violet: reconstructed field
the physics supplies everything between the sensors

What to Report for a Turbulent Flow

THE TWO NUMBERS AN ENGINEER ASKS FOR

$$St = \frac{fD}{U}, \quad C_D = \frac{F_D}{\frac{1}{2}\rho U^2 D}$$

DRAW COEFFICIENT

Integrate pressure and shear over the obstacle. Highly sensitive to the near-wall solution, so it is the first quantity to expose an unresolved boundary layer.

STROUHAL NUMBER

The shedding frequency, non-dimensionalised. For a cylinder it sits near 0.2 across a wide Re range — a strong and easily checked target.

PRESSURE DROP

Across a pipe section with an obstacle. This is what sizes a pump, and it is the deliverable in most industrial cases.

MEAN PROFILES

Velocity across the channel, compared with the log law where applicable. Checks the wall treatment rather than the bulk flow.

None of these is a contour plot. A turbulent result is judged on integrated quantities, because those are what survive the averaging.

Where This Stops Working

Raw N-S at high Re

No closure, no averaging — just Navier–Stokes at $Re = 10^5$ and hope.

Result: a smooth, plausible, wrong field. The most dangerous failure here.

Separation and recirculation

Eddy-viscosity closures are calibrated on attached flows and degrade behind bluff bodies.

Symptom: wake length wrong by tens of percent. Affects classical codes equally.

Near-wall resolution

The viscous sublayer thins as Re grows and needs the densest sampling of anywhere.

Fix: wall functions, or accept that drag will be underestimated.

Unsteady versus averaged

RANS predicts a mean. Asking it for vortex shedding is a category error.

Fix: URANS or a scale-resolving approach — and much more compute.

Validation gap

A PINN turbulence result has no grid-convergence study to stand behind it.

Fix: compare against benchmark data. Never present it as a certified answer.

A student who can name these five is better prepared than one who has produced a pretty wake plot.

Questions

TEN TO TEST WHETHER TODAY LANDED

- why does the point count grow as Re to the nine-fourths?
- which term survives averaging, and why that one?
- state the closure problem in one sentence
- why does deriving an equation for the Reynolds stress not help?
- what does an eddy viscosity assume about the turbulence?
- why must the eddy viscosity be constrained non-negative?

AND FOUR MORE

seven

what does a wall function buy a PINN specifically?

eight

prescribe a closure, or learn one - and when?

nine

why is a smooth network biased towards the wrong answer?

ten

why is data assimilation the honest case rather than solving?

THE ONE WITH NO SETTLED ANSWER

Can a learned closure generalise beyond the flows it was trained on?

BEFORE L10.1

Ex_09.2 notebooks 01 and 02. Bring the eddy-viscosity profile.

Ex_9.2 — Obstacle in a Pipe, Your Choice of Shape

TASKS

- 1 Choose an obstacle shape — cylinder, square, ellipse, diamond or aerofoil — and set the channel geometry.
- 2 Solve the steady mean flow at moderate Re with an effective viscosity, and verify against the inlet profile.
- 3 Sweep the Reynolds number and record drag, pressure drop and, where shedding appears, the Strouhal number.
- 4 Compare two shapes at identical Re and blockage, and explain the difference in drag physically.
- 5 Push Re until the solution degrades, and describe how the failure appears.

QUESTIONS — AND WHERE THIS DECK ANSWERS THEM

Why does averaging create more unknowns than equations? sl. 5, 6

What does the eddy-viscosity hypothesis assume, and when does it fail? sl. 7, 12

Why is the near-wall region the expensive part? sl. 8

Why report drag and Strouhal rather than a contour plot? sl. 11

Why is data assimilation the honest use case? sl. 10

Same control panel as Ex_9.1, with geometry added. The report is generated for you; the interpretation is not.

The L13 MiniProject: Turbulence as Open Research

- ▶ Everything in this lecture stops at prescribed closures. The open question — can a network learn the closure itself — is a live research topic, and it is the right shape for a MiniProject.

- | | | |
|----------|-------------------------------|--|
| A | Learn the closure | Represent v_t by a second network and train it against RANS plus reference data. Does it generalise to a Reynolds number it never saw? |
| B | Assimilate sparse data | Reconstruct a mean field from a handful of virtual sensors. How few can you use before the reconstruction fails, and where does it fail first? |
| C | Enforce the wall law | Impose the logarithmic profile as a hard constraint instead of resolving the sublayer. How much accuracy does it buy per unit of compute? |
| D | Map the limit | Chart where a plain PINN stops working across Re , geometry and sampling. A negative result, carefully established, is a real contribution. |

Track D is deliberately included: establishing where a method fails is worth as much as making it succeed, and it is honest work.

Key Takeaways

- 1 Turbulence is modelled, not solved** Resolving every scale costs Re^3 points. Nobody does it for engineering flows — not with finite volumes, and not with a network.
- 2 Averaging creates the closure gap** Reynolds averaging is exact, but the nonlinear term leaves six unknown stresses and no equations for them. The hierarchy never closes.
- 3 Eddy viscosity makes it tractable** Assume the Reynolds stress acts like viscosity and you recover the L9.1 equations with ν replaced by $\nu + \nu_t$. Same code, new coefficient.
- 4 The wall is the expensive part** The sublayer thins as Re grows. The log law is universal, so impose it rather than resolve it.
- 5 Assimilation is the honest win** Reconstructing a field from sparse sensors under a physical constraint is something no classical solver does. That is where a PINN belongs.

Next — L10: Electrochemical Devices, where a parabolic transport equation couples to an elliptic potential field.